

Spectral-Local Residual Networks for Wave Inverse Problems

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Abstract

TODO

1 Introduction

Problems such as phase retrieval, inverse scattering, and forward PDE prediction combine long-range coupling with localized structure. In lensless computed imaging, for example, each pixel in the measured image depends on the entirety of the target object, while at the same time it is essential to process local textural information (i.e. speckle). (cites)

Similarly, in fluid dynamics large-scale coherent dynamics coexist with localized features such as shocks and vortices. Consequently, classical numerical PDE solvers require fine discretization (for forward-solvers) or iterative convex optimization (for inverse-solvers). These techniques are highly computationally expensive.

Standard computer vision architectures adapted to physics-prediction tasks provide fast approximate solutions (TODO say how much faster, motivate why this is important) but are typically too inaccurate to replace conventional numerical solvers in domains with prominent scale-mixing (cites). The accuracy gap traces to specific architectural mismatches: architectures that are efficient on natural images, such as CNNs, lack global receptive fields and are consequently poorly suited to modeling long-range interactions.

This is a direct contrast to Fourier neural operator (FNO) networks, which, while highly suited to modeling continuous structure in physical systems, rely on spectral truncation and thus cannot represent sharp features (cites).

(TODO maybe explain why we're not considering diffusion or transformer / attention models (basically they're too slow, too data hungry)). (TODO transition to instruding FNO, explaing why it's a strong class of models).

Recent extensions of the FNO architecture mitigate these limitations by coupling global Fourier layers with spatially-local operators [4, 3]. These convolutional, U-Net, or localized-kernel branches provide a complementary inductive bias for fine-scale structure while preserving the long-range mixing of Fourier layers. Such hybrids improve some PDE benchmarks, but they are not uniformly better than strong spectral-only architectures such as the factorized Fourier neural operator (cite). (cites)

Coherent diffractive imaging and phase retrieval is a scientifically interesting (TODO why?) task that reveals the failure modes of existing approaches because it requires ingestion *and* synthesis of both high- and low-spatial frequency information. Our experiments demonstrate that FNO-based architectures that are strong baselines (or SOTA-class) on difficult forward-prediction tasks such as compressible Navier-Stokes fail entirely to predict real-space images from diffraction measurements

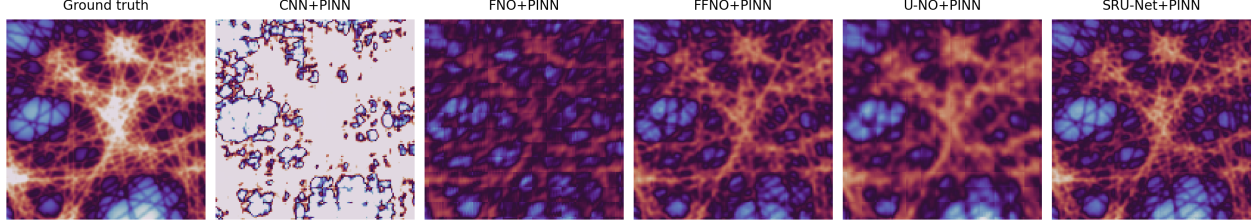


Figure 1: **Phase reconstruction comparison on synthetically-generated diffraction data.** The cropped comparison shows ground truth, CNN+PINN, FNO+PINN, FFNO+PINN, U-NO+PINN, and SRU-Net+PINN under the same $N = 128$ diffraction data protocol. Each prediction is aligned to the ground-truth phase by a single global circular offset before display.

(cites or reference to relevant fig in this manuscript), though they still outperform convolutional networks.

(TODO maybe explain why we’re not considering diffusion or transformer / attention models (basically they’re too slow, too data hungry)).

Our contributions in this paper are twofold. First we introduce SRU-Net, a computationally-efficient spectral-convolutional architecture that yields better performance on coherent diffractive imaging than existing neural operator and hybrid architectures. Second, we propose a physics-consistency training objective that significantly improves prediction accuracy, regardless of model class.

(TODO summarize the architecture, introduce the other benchmarks, enumerate the baselines, etc.)

Our primary benchmark uses synthetically generated CDI diffraction data. We compare SRU-Net with matched CNN, FNO, FFNO, and U-NO baselines trained through the same objective. We include ablations that vary the network’s topology, connectivity, and the placement of spectral and convolutional components. These experiments confirm the importance of routing residual connections outside layer activations, as discovered by (cite FFNO authors), and illustrate the distinct roles of the inverse-mapping and image synthesis components of the overall network.

We further test SRU-Net on a Born–Rytov diffraction tomography (BRDT) inverse-scattering task of recovering a spatial scattering potential from wave measurements. This evaluation provides a second wave-propagation inverse problem setting beyond CDI. Under the same supervised object loss plus Born-consistency loss, SRU-Net gives lower image-space error than FFNO while retaining sharper object boundaries. Spectral mixing therefore helps interpret globally coupled wave measurements, but a real-space synthesis path remains important for recovering localized geometry. (TODO does this last sentence belong somewhere else?)

Third, we evaluate SRU-Net on PDEBench compressible Navier–Stokes. It outperforms standard PDEBench baselines but falls short of FFNO. FFNO’s much larger depth (24 spectral layers vs. SRU-Net’s 4) represents a scaling dimension that we leave unexplored because it comes at the cost of reduced inference throughput. [6]

(TODO should there be an equal-network depth comparison? i.e. SRUNet vs. FFNO with 4 layers instead of 24)

2 Related Work

2.1 Operator learning for wave and PDE field maps

Neural operators learn mappings between fields and have become a central model class for scientific machine learning. Fourier neural operators (FNOs) apply global spectral convolutions to represent solution operators for parametric PDEs [2]. Factorized Fourier neural operators (FFNOs) make this parameterization substantially deeper and more efficient by factorizing Fourier transforms, sharing the kernel integral operator across layers, and using residual temporal updates [5]. This combination gives FFNO reported state-of-the-art-class performance on PDE benchmarks, including Navier-Stokes-style settings, which makes it a strong baseline rather than only an efficient FNO variant. The global receptive field of these models is attractive for wave propagation, diffraction, and fluid dynamics, where distant regions of a field are coupled by the governing physics. PDEBench provides a controlled setting for evaluating such models on forward physical prediction tasks, including compressible Navier-Stokes dynamics [6].

The same global-coupling argument motivates operator-style models for wave inverse problems. In diffraction imaging, local object structure is observed through global Fourier measurements; in compressible flow, local shocks and vortices interact with coherent large-scale dynamics. Pure spectral models can represent global coupling directly, but Fourier-mode truncation can bias predictions toward lower-frequency structure. The central architectural problem is therefore not only how to model field-wide interactions, but how to retain localized detail while doing so.

2.2 Physics-constrained wave inverse reconstruction

Hoidn, Mishra, and Mehta introduced PtychoPINN for unsupervised scanning-CDI reconstruction [1]. The method learns an inverse map from diffraction data to a complex real-space object, then trains it only after passing the predicted object through known coherent diffractive imaging physics: overlap-based real-space consistency, the probe interaction, and far-field diffraction. Its loss is defined in photon-count space with a Poisson model, so the network is optimized against the measured diffraction statistics rather than paired ground-truth objects.

This paper uses that physics-constrained formulation as the primary evaluation setting for a broader wave inverse architecture. PtychoPINN supplies the measurement-consistency objective for CDI, but the model contribution is not a new coherent diffractive imaging forward model. The question is whether the reconstruction network used inside a wave inverse problem should remain a local CNN-style image model, or whether it should combine local spatial processing with explicit spectral mixing.

2.3 Spectral-local architectures

Several recent architectures combine spectral mixing with local or multiscale image structure. U-FNO-style models place Fourier layers within a U-Net-like multiresolution architecture [4], while localized-kernel neural operators strengthen the local component of operator models [3]. These approaches reflect a common design principle: spectral layers are useful for field-wide coupling, but local convolutional and residual paths remain important for high-frequency detail and spatially localized structure.

SRU-Net follows this principle in a network designed first for wave inverse reconstruction. It combines spectral-local encoder blocks, residual refinement, and decoder-side upsampling as a single reconstruction architecture trained through a physics-consistency loss. The PDEBench CNS experiments then test the same architecture in a supervised forward-prediction setting, where

FNO and FFNO are natural operator-learning baselines and where SRU-Net should be interpreted as competitive transfer evidence rather than as the central benchmark result.

3 Methods

3.1 Task-agnostic network formulation

Let x denote the task input and y the target field or reconstruction. We write the learned network as

$$\hat{y} = G_\theta(x). \quad (1)$$

For CDI, x is a stack of diffraction measurements and $\hat{y}$ is a complex object-patch representation, meaning local crops of the reconstructed complex object represented by real and imaginary channels. For BRDT, x is a Born-derived image-space initialization and $\hat{y}$ is the recovered scattering potential. For PDEBench CNS, x is a temporal history of physical fields and $\hat{y}$ is the predicted next state. The tasks use different input representations and objective functions, but share the same architectural principle: G_θ combines global spectral mixing with local convolutional feature recovery.

We decompose the network as

$$G_\theta = D_\theta \circ B_\theta \circ E_\theta, \quad (2)$$

where E_θ is the spectral-local encoder, B_θ is a ResNet bottleneck at reduced spatial resolution, and D_θ is a convolutional upsampling decoder.

We use subscripts to distinguish architectural positions rather than tensor dimensions. The index ℓ denotes the encoder block that maps z_ℓ to $z_{\ell+1}$; j indexes the six residual bottleneck blocks; s indexes resolution levels used for encoder-decoder skip connections; and k indexes Fourier modes. The retained Fourier-mode set is K , with $m = |K|$ in the architecture diagram. Superscripts such as $B_j^{\times 6}$ denote repeated blocks rather than exponentiation. Task subscripts such as CDI and CNS label task-specific outputs.

3.2 Spectral-local encoding

For a feature tensor z_ℓ , a spectral-local encoder block has the form

$$z_{\ell+1} = \sigma \left(L_\ell z_\ell + \mathcal{F}^{-1} [R_\ell(k) \mathcal{T}_K(\mathcal{F}(z_\ell))(k)] \right), \quad (3)$$

where L_ℓ is a learned local convolutional map, $\mathcal{F}$ is the 2D Fourier transform, $\mathcal{T}_K$ retains a fixed set of Fourier modes, $R_\ell(k)$ is a learned complex spectral mixing tensor over retained modes, and σ is the GELU nonlinearity used in the implemented SRU-Net. In residual form, this can be written as

$$z_{\ell+1} = z_\ell + \sigma \left(L_\ell z_\ell + S_\ell z_\ell \right), \quad (4)$$

with S_ℓ denoting the truncated spectral operator. The local path preserves high-frequency and edge-like information that may be weakened by spectral truncation, while the spectral path provides global coupling.

3.3 Spectral-convolutional residual architecture

The implemented SRU-Net consists of an initial convolution, four spectral-local encoder blocks, two downsampling stages, a six-block residual bottleneck at $N/4$ spatial resolution, and a convolutional upsampling decoder. Figure 2 summarizes the data flow, and Table 1 lists the architectural choices that determine the shape of the model.

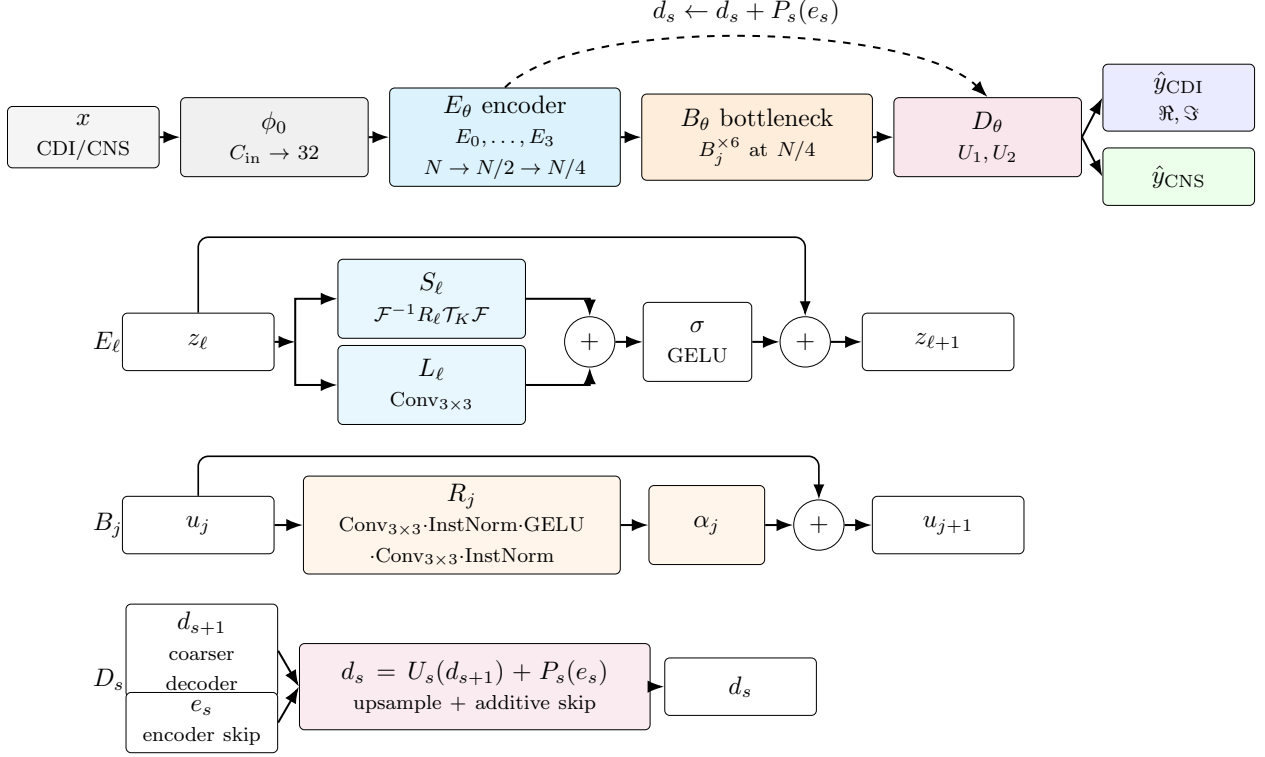


Figure 2: **SRU-Net combines spectral global mixing with local spatial processing.** Top: x is the task input, ϕ_0 projects C_{in} input channels to width 32, E_θ is the encoder over resolutions $N \rightarrow N/2 \rightarrow N/4$, B_θ is the six-block bottleneck, D_θ is the decoder, and $\hat{y}_{\text{CDI}}, \hat{y}_{\text{CNS}}$ are task-specific outputs. Middle: encoder block E_ℓ maps z_ℓ to $z_{\ell+1}$ using spectral path $S_\ell = \mathcal{F}^{-1} R_\ell \mathcal{T}_K \mathcal{F}$, local convolution path L_ℓ , GELU nonlinearity σ , and a residual bypass. The bottleneck detail shows that bottleneck block B_j maps u_j to u_{j+1} by applying R_j , then scaling it by α_j before adding it to the bypassed input. The R_j box uses centered dots to list left-to-right sequential application; $\text{Conv}_{3 \times 3}$ denotes a 3-by-3 convolution and InstNorm denotes instance normalization. The decoder detail writes the stage directly as $d_s = U_s(d_{s+1}) + P_s(e_s)$, where U_s upsamples the coarser decoder feature and P_s projects the encoder skip feature before additive fusion.

In the architecture diagram, ϕ_0 is the initial channel projection, E_0 through E_3 are encoder blocks, B_j denotes a residual bottleneck block, and U_1, U_2 are decoder upsampling stages. The spatial resolution is denoted by N , C_{in} is the task-specific input channel count, m is the number of retained Fourier modes, and e_s, d_s are encoder and decoder features at scale s . The skip projection P_s maps encoder features into the decoder channel space before additive fusion. Within encoder block E_ℓ , z_ℓ is the input feature tensor, $S_\ell = \mathcal{F}^{-1} R_\ell \mathcal{T}_K \mathcal{F}$ is the truncated spectral operator, L_ℓ is the local convolutional path, and σ is GELU. Here $\mathcal{F}$ is the Fourier transform, $\mathcal{T}_K$ keeps the retained modes, and R_ℓ mixes those modes. Within bottleneck block B_j , u_j is the bottleneck feature tensor, $R_j = \text{Conv}_{3 \times 3} \cdot \text{InstNorm} \cdot \text{GELU} \cdot \text{Conv}_{3 \times 3} \cdot \text{InstNorm}$ is applied before the residual addition, and α_j is its learned residual scale; the centered dot lists operations in left-to-right order. Here $\text{Conv}_{3 \times 3}$ denotes a 3-by-3 convolution. Within decoder stage D_s , U_s upsamples the coarser decoder state and P_s projects the matching encoder feature before fusion.

When skip connections are enabled, encoder features e_s are fused into decoder states d_s . For

Table 1: **SRU-Net architecture summary.** The same architecture is used for CDI reconstruction and CNS field prediction, with task-specific input representations, output variables, and objective functions.

Component	Implementation details
Input representation	CDI uses grouped object-patch channels and returns complex real/imaginary patches. CNS uses real-valued field histories as input and predicts the next density, velocity, and pressure fields. In both settings, the first convolution maps the input to width 32.
Encoder	Four spectral-local blocks. Each block adds a learned Fourier-mode mixing branch and a reflected-padding 3×3 convolution branch inside an outer residual update; the branch sum is passed through GELU before the outer residual addition. The retained-mode count m is a configurable parameter; the reported base setting uses $m = 12$ unless otherwise stated. The encoder downsamples after the first two blocks, giving $N \rightarrow N/2 \rightarrow N/4$ and widths 32, 64, and 128.
Bottleneck	Six constant-resolution residual blocks at $N/4$. Each block uses a reflected-padding 3×3 convolution, instance normalization, GELU, a second reflected-padding 3×3 convolution, instance normalization, and a learned residual scale. The spectral-residual variant replaces this local bottleneck with residual blocks that add a shared factorized spectral branch.
Decoder and skips	The CDI path uses two transposed-convolution upsampling stages. The selected CNS row uses two pixel-shuffle upsampling stages and projected additive encoder-decoder skips from the $N/2$ and N encoder resolutions.
Output and loss	CDI predicts real/imaginary complex patches and trains through the coherent diffractive imaging consistency and diffraction operators. BRDT predicts a real scattering potential and trains with supervised object loss plus Born consistency. CNS predicts real field channels and trains against the next physical state.

additive fusion,

$$d_s = U_s(d_{s+1}) + P_s(e_s), \quad (5)$$

where U_s upsamples the coarser decoder state and P_s projects the encoder feature into the decoder channel space. For concatenative fusion,

$$d_s = P_s([U_s(d_{s+1}), e_s]). \quad (6)$$

The CDI network outputs complex object patches in real/imaginary format. In BRDT, the same reconstruction architecture outputs a single real scattering potential. In the PDEBench experiments, the architecture is evaluated on real-valued physical field histories and next-state targets rather than diffraction measurements.

3.4 Physics-consistency loss for CDI

For CDI, SRU-Net predicts local complex object patches,

$$\hat{o} = G_\theta(x). \quad (7)$$

The prediction is passed through a real-space consistency operator F_c , which stitches overlapping local patches into a common object canvas and re-extracts position-consistent patches. The consistent object patches are then passed through a known diffraction map F_d :

$$\hat{x} = F_d(F_c(\hat{o}; r); P), \quad (8)$$

where r denotes scan positions and P is the probe. The CDI objective is defined in measurement space:

$$\mathcal{L}_{\text{CDI}} = \mathcal{D}(x, F_d(F_c(G_\theta(x); r); P)) + \lambda R(G_\theta(x)). \quad (9)$$

Here $\mathcal{D}$ is the experiment-specific measurement loss, such as Poisson negative log likelihood or amplitude/intensity reconstruction error, and R denotes optional reconstruction regularization. This preserves the physics-informed training principle: the network is optimized through the forward diffraction model rather than only through a supervised object-target loss.

3.5 Born–Rytov diffraction tomography

We also evaluate SRU-Net on a Born–Rytov diffraction tomography (BRDT) inverse-scattering task. The target is the physical scattering potential

$$q(x, z) = k_m^2 \left[\left(\frac{n(x, z)}{n_m} \right)^2 - 1 \right], \quad (10)$$

where $n(x, z)$ is the refractive-index field, n_m is the background medium index, and k_m is the in-medium wavenumber. Under the first Born approximation, each incident plane wave produces a scattered field whose far-field samples are linear measurements of the Fourier transform of q on the corresponding Ewald arc. We write this discrete forward map as

$$s = \mathcal{B}(q), \quad (11)$$

where $s \in \mathbb{R}^{A \times D \times 2}$ is the complex sinogram over illumination angles and detector frequencies, represented by real and imaginary channels.

The BRDT network input is a Born-derived image-space initialization x_{BRDT} computed from the measured sinogram. The model predicts a single-channel scattering potential,

$$\hat{q}_\theta = G_\theta(x_{\text{BRDT}}). \quad (12)$$

All neural rows use the same input representation and the same differentiable Born forward operator during training.

The BRDT objective combines supervised reconstruction with measurement-space Born consistency:

$$\begin{aligned} \mathcal{L}_{\text{BRDT}} = & \lambda_{\text{img}} \|\hat{q}_\theta^{\text{norm}} - q^{\text{norm}}\|_1 + \lambda_{\text{abs}} \|\mathcal{B}(\hat{q}_\theta) - s\|_1 \\ & + \lambda_{\text{rel}} \frac{\|\mathcal{B}(\hat{q}_\theta) - s\|_2}{\|s\|_2 + \epsilon} + \lambda_{\text{tv}} \text{TV}(\hat{q}_\theta) + \lambda_{\text{pos}} \|\min(\hat{q}_\theta, 0)\|_2^2. \end{aligned} \quad (13)$$

The image loss is computed in the normalized target space used by the model. Before applying the Born operator, predictions are converted back to physical q units using training-set normalization statistics. This prevents the measurement-consistency term from being evaluated on normalized, dimensionless outputs. We use $\lambda_{\text{img}} = 1$, $\lambda_{\text{abs}} = 0.1$, $\lambda_{\text{rel}} = 0.1$, $\lambda_{\text{tv}} = 10^{-5}$, and $\lambda_{\text{pos}} = 10^{-4}$.

3.6 PDEBench CNS supervised prediction

For the CNS benchmark, the same architecture is evaluated as a supervised forward predictor. Let

$$u_t = [\rho_t, v_{x,t}, v_{y,t}, p_t] \quad (14)$$

be the density, velocity, and pressure state. Given a history length h , the input is

$$x_t = [u_{t-h}, \dots, u_{t-1}], \quad (15)$$

and the model predicts

$$\hat{u}_t = G_\theta(x_t). \quad (16)$$

The training objective is mean-squared error on normalized physical fields:

$$\mathcal{L}_{\text{CNS}} = \|\hat{u}_t - u_t\|_2^2. \quad (17)$$

Evaluation is performed after denormalization and reports root mean squared error (RMSE), normalized RMSE (nRMSE), relative L_2 , and Fourier-band RMSE over low, mid, and high radial-frequency bands. The high-frequency metric is included because shocks and localized vortical features are precisely where global spectral models may lose detail.

4 Experiments

4.1 Primary CDI benchmark

The primary benchmark is a 128-by-128 CDI reconstruction task with synthetically-generated diffraction data. Each sample is generated from sharp line-like objects, a known probe, and simulated far-field diffraction measurements. This benchmark is the direct test of the physics-informed reconstruction model: the network predicts complex object patches, the CDI consistency and diffraction operators map the prediction back to detector space, and reconstruction quality is measured against held-out object amplitude and phase.

The synthetic CDI comparison uses $N = 128$ object grids, two training objects, two held-out test objects, 10^9 photons per exposure, 40 training epochs, learning rate 2×10^{-4} , ReduceLROnPlateau scheduling, and complex-valued real/imaginary network output.

CDI results are reported by model and training procedure: SRU-Net + PINN, FNO + PINN, and CNN + PINN denote architectures trained through the CDI diffraction forward model rather than through direct supervised object targets. The current FFNO rows are labeled FFNO-local proxy because they used post-stack local CNN refinement; no-refiner FFNO rows are being rerun before canonical FFNO claims are made. Rows without “+ PINN” denote non-PINN baselines. Appendix Table 8 reports the effective configuration of each reported model row; parameter counts are unique trainable parameters in the effective prediction model.

4.2 Secondary inverse-scattering benchmark: BRDT

The BRDT experiment is a controlled secondary inverse-scattering study. It uses 128×128 scattering-potential fields and simulated complex sinograms from a differentiable Born forward model with 64 illumination angles and 128 detector samples. We use it to test whether the CDI reconstruction architecture transfers to a related wave inverse problem; it is not used as a primary benchmark claim. We train SRU-Net and FFNO for 40 epochs on the same 2048/256/256 train/validation/test split, using Adam with learning rate 2×10^{-4} , batch size 16, seed 42, and ReduceLROnPlateau scheduling. Both rows use the same Born-derived image initialization as input and the same supervised-plus-Born-consistency objective defined above.

Table 2: **Synthetic CDI model comparison under physics-informed training.** All rows use the same $N = 128$ line-pattern protocol and CDI physics-informed objective.

Model	Amp MAE ↓	Phase MAE ↓	Amp MSE ↓	Phase MSE ↓	Amp SSIM ↑	Phase SSIM ↑
CNN	0.1232	0.7863	0.0237	1.1843	0.7326	0.2638
FNO	0.1248	0.1435	0.0240	0.0323	0.7409	0.9335
FFNO-local proxy	0.0628	0.0828	0.0062	0.0109	0.9348	0.9816
SRU-Net	0.0269	0.0721	0.0012	0.0078	0.9881	0.9947
U-NO	0.0932	0.0683	0.0132	0.0073	0.8280	0.9569

4.3 Secondary cross-domain benchmark: PDEBench CNS

The secondary benchmark is PDEBench 2D compressible Navier-Stokes at native 128×128 resolution. It stresses the same architectural tradeoff in a different physical setting: coherent global flow and wave structure must be predicted while preserving localized shocks, vortices, and high-gradient field features. It therefore tests whether the SRU-Net architecture transfers beyond the CDI inverse problem.

The selected CNS data are the official 128-by-128 periodic compressible-flow training trajectories at Mach number $M = 1.0$, shear viscosity $\eta = 0.01$, and bulk viscosity $\zeta = 0.01$. The spatial grid has $\Delta x = \Delta y = 1/128 = 0.0078125$, the temporal spacing is $\Delta t = 0.05$, and the spatial boundary condition is periodic, meaning that values wrap around at the domain edges as on a torus. Each trajectory contains the fields density, v_x , v_y , and pressure on a 128×128 grid. Splits are trajectory-level, normalization is computed from the train split only, and metrics are evaluated on denormalized predictions.

5 Results

The CDI results evaluate wave inverse reconstruction quality on synthetically-generated diffraction data. The BRDT results test the same reconstruction architecture on inverse scattering. The CNS results evaluate cross-domain next-state prediction on compressible flow through a secondary benchmark table and targeted architecture ablations.

5.1 CDI reconstruction and architecture ablations

The synthetic CDI benchmark compares local, spectral, and hybrid spectral-convolutional networks under the same synthetically-generated diffraction data protocol. Figure 1 shows the matched phase-crop comparison. Table 2 reports the same-objective PINN model comparison. Table 3 reports paired PINN versus supervised controls for architectures where both rows are available.

The synthetic CDI benchmark shows a clear separation between purely local, purely spectral, and hybrid spectral-convolutional models. SRU-Net gives the best amplitude-side reconstruction among the PINN-trained models by MAE, MSE, and SSIM. U-NO gives the lowest phase MAE and MSE, while SRU-Net gives the strongest phase SSIM. The historical FFNO-local proxy improves strongly over the CNN and FNO PINN models but does not match SRU-Net’s amplitude-side reconstruction on this CDI task. The supervised FFNO control improves phase MAE but collapses amplitude quality relative to the FFNO-local proxy + PINN row. This historical objective-control comparison will be replaced by the no-refiner FFNO reruns before being used as canonical pure-FFNO evidence. The same pattern appears for U-NO: supervised training improves phase MAE, while PINN training gives stronger amplitude reconstruction and phase SSIM.

Table 3: **Synthetic CDI objective controls.** Each block compares physics-informed and supervised training for the same architecture where both rows are available. Lower MAE is better; higher SSIM is better.

Objective	Amp MAE	Phase MAE	Amp SSIM	Phase SSIM
<i>CNN</i>				
PINN	0.1232	0.7863	0.7326	0.2638
Supervised	0.4027	0.7862	0.2772	0.4964
<i>FFNO-local proxy</i>				
PINN	0.0628	0.0828	0.9348	0.9816
Supervised	0.3864	0.0466	0.2484	0.9372
<i>U-NO</i>				
PINN	0.0932	0.0683	0.8280	0.9569
Supervised	0.3207	0.0563	0.2689	0.9105

Table 4: **CDI spectral-component ablations.** Rows test whether SRU-Net’s performance is explained by spectral capacity alone or by the placement of spectral mixing inside an architecture with local convolutional paths, residual refinement, and convolutional upsampling. Lower MAE is better.

Row	Changed component	Amp MAE	Phase MAE
SRU-Net	reference	0.0269	0.0721
SRU-Net with FFNO-style bottleneck	replace ResNet bottleneck	0.0312	0.0876
Spectral-bottleneck SRU-Net	reference spectral-bottleneck model	0.0249	0.0929
Spectral-bottleneck SRU-Net, one downsampling stage	shallower encoder	0.0355	0.0644
Spectral-bottleneck SRU-Net, linear decoder	lighter decoder	0.1347	0.1169

Spectral-component ablations. Table 4 tests whether SRU-Net’s CDI advantage is explained by simply adding or moving spectral components. Replacing the SRU-Net bottleneck with an FFNO-style bottleneck worsened both amplitude and phase error. A lighter linear decoder for the spectral-bottleneck model regressed sharply in amplitude quality, while reducing the spectral-bottleneck model to one downsampling stage improved phase MAE but lost amplitude fidelity. Spectral coupling was most effective when paired with local convolutional paths, residual refinement, and convolutional upsampling; adding or moving spectral components alone did not reproduce the SRU-Net result.

5.2 PDEBench CNS results

Table 5 reports a single matched-condition CNS headline ranking. All four rows share the same fixed contract: `history_len=5`, 512/64/64 train/validation/test trajectories, 40 training epochs, MSE loss, and identical optimizer and normalization settings. FFNO gives the lowest aggregate and Fourier-band errors. SRU-Net* is the next strongest row by aggregate error and by high-frequency `fRMSE`, with FNO third and U-Net well behind on this capped slice. Because this table uses a capped 512/64/64 split, we treat it as a matched-condition architecture comparison rather than a full-training PDEBench benchmark.

The locked `history_len=2`, 2048/256/256 capped CNS authority recorded in `pdebench.cns_paper.2048cap-ex` is preserved as larger-cap bounded context only and is no longer the headline ranking source.

The CNS matched-condition headline table uses 512 training trajectories as a secondary architecture comparison. The asterisk marks the CNS spectral-bottleneck row.

Table 5: **PDEBench CNS matched-condition results.** Single matched-condition headline table. All rows share the `history_len=5`, 512/64/64 train/validation/test split, 40-epoch capped contract used for the head-to-head ranking. SRU-Net* denotes the spectral-bottleneck SRU-Net variant. Low-, mid-, and high-band RMSE are Fourier-band errors. Lower values are better. Model sizes and row configurations are reported in Appendix Table 8.

Model	nRMSE ↓	Low RMSE ↓	Mid RMSE ↓	High RMSE ↓
FFNO	0.0198	1.130	0.042	0.102
SRU-Net*	0.0331	1.854	0.171	0.262
FNO	0.0384	2.134	0.122	0.433
U-Net	0.5386	30.988	0.645	1.743

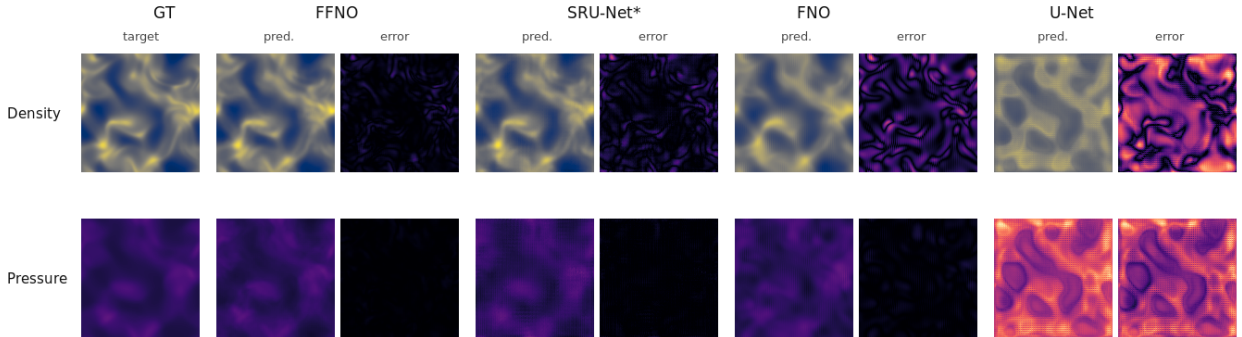


Figure 3: **PDEBench CNS fixed-sample predictions and errors (adjacent context).** The shared test sample comes from the earlier 2-input-time-step 512/64/64 CNS visual comparison and is shown as adjacent-context illustration of the per-row error structure. The matched-condition headline ranking in Table 5 uses the `history_len=5` contract; this figure is not the same-condition fixed-sample comparison for that contract.

5.3 Secondary BRDT inverse-scattering experiment

BRDT provides a controlled inverse-scattering test in which the target is the physical scattering potential q and the input is a Born-derived initialization from simulated wave measurements. The current historical 40-epoch bundle compares SRU-Net with an FFNO-local-refiner proxy under the same supervised object loss plus Born-consistency objective. SRU-Net gives the strongest image-space reconstruction in that bundle. A corrected no-refiner BRDT FFNO rerun is pending, so the proxy row is not used as a pure FFNO-paper-stack claim. Vanilla FNO is not included as a main BRDT comparison row because it did not converge under this supervised-plus-Born objective. An objective-control visualization in Appendix Fig. 5 shows that the same FNO architecture recovers meaningful structure under a physics-only loss, so the failure is tied to the training objective rather than presented as a controlled method comparison in the main BRDT figure.

Among the reported 40-epoch neural rows, SRU-Net gives the lower image-space error on this BRDT split, with relative $L_2(q)$ of 0.2875 versus 0.3324 for the historical FFNO-local-refiner proxy. The model-based Born inverse provides a non-neural reference. Both neural rows evaluate in under 0.7 seconds over the 256-sample test split, roughly an order of magnitude faster than the iterative model-based inverse. These results support the transfer behavior of SRU-Net, while the paper’s main quantitative claims remain the CDI and CNS benchmarks.

Table 6: **Controlled BRDT inverse-scattering study.** SRU-Net and the historical FFNO-local-refiner proxy are trained for 40 epochs with the same supervised object loss plus Born-consistency loss. The model-based Born inverse is a non-neural reference. This secondary study is reported as transfer context rather than as a primary benchmark. Throughput is measured over the 256-sample test split on the same GPU. Lower error is better; higher PSNR, SSIM, and throughput are better. Parameter counts are reported for learned neural rows; exact adapter settings are reported in Appendix Table 8.

Model	Procedure	Rel. $L_2(q)$	PSNR	SSIM	Params	Samples/s
Model-based Born inverse	iterative inverse	0.3796	28.23	0.9201	–	29.6
SRU-Net	supervised + Born	0.2875	30.64	0.9556	142k	385.3
FFNO-local proxy	supervised + Born	0.3324	29.38	0.9455	36.7k	375.5

Table 7: **Selected SRU-Net* follow-up comparisons on PDEBench CNS.** SRU-Net* denotes the spectral-bottleneck SRU-Net variant. The ablation axis is the architectural or input-context change named in the first column; the second column records run context only. All rows use 40 training epochs. Low-, mid-, and high-band RMSE are Fourier-band errors. Lower values are better.

Tested axis	Run context	nRMSE ↓	Low RMSE ↓	Mid RMSE ↓	High RMSE ↓
Reference setting	512/64/64 split; 2 input steps	0.062	3.476	0.280	0.435
Longer temporal context	512/64/64 split; 3 input steps	0.046	2.560	0.216	0.347
More retained Fourier modes	512/64/64 split; 2 input steps	0.065	3.676	0.242	0.347
Deeper shared-weight spectral bottleneck	1024/128/128 split; 2 input steps	0.045	2.483	0.217	0.294

5.4 PDEBench CNS ablations

Table 7 summarizes selected SRU-Net* ablations on the CNS benchmark. The rows separate the tested architectural or input-context axis from the run context, showing how temporal context, spectral capacity, and bottleneck depth affect aggregate and Fourier-band errors. All rows in this summary use a 40-epoch training budget; longer convergence follow-ups are kept out of the ablation table.

6 Discussion

The core architectural claim is that wave inverse reconstruction benefits from networks that encode both propagation-scale coupling and local spatial features. The task physics should constrain the output through a differentiable forward model, while the reconstruction network should match the structure of the inverse map. In CDI, diffraction couples the object through Fourier-domain wave physics, while the object itself contains sharp local features. In compressible flow, used here as a secondary transfer setting, large-scale coherent dynamics coexist with localized shocks and vortices.

The SRU-Net encodes this global-local structure directly. Fourier-style encoder blocks provide global mixing; local convolutional branches and decoder features preserve high-frequency information; the ResNet bottleneck gives the model capacity to refine features at reduced spatial resolution. This differs from both pure CNN reconstruction and pure spectral operator modeling.

The CDI architecture table and spectral-component ablations point to the same conclusion: SRU-Net’s performance depends on where spectral coupling is introduced and how it is paired

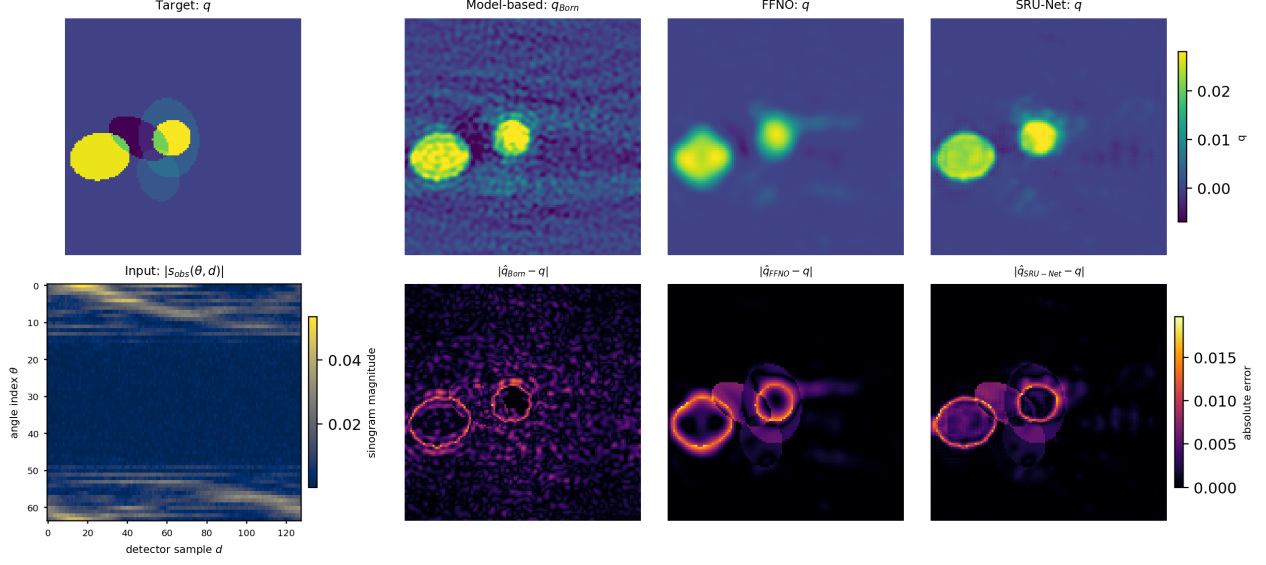


Figure 4: **BRDT representative example.** The figure separates the BRDT input $|s_{\text{obs}}(\theta, d)|$ from the target scattering potential q . The top row shows the target and three estimates $\hat{q}$; the bottom row shows the input sinogram magnitude followed by absolute target-domain errors under a shared error scale. The FFNO-labeled panel is the historical local-refiner proxy, not the pending corrected no-refiner BRDT FFNO rerun.

with local convolutional paths, residual refinement, and decoder-side upsampling. A remaining encoder-branch ablation will remove one path at a time under the same CDI dataset, training, and evaluation protocol.

The evidence also argues for a narrow interpretation. The primary claim is about wave inverse reconstruction: the CDI result compares local CNN, FNO, FFNO, SRU-Net, and spectral SRU-Net rows under one synthetically-generated diffraction data protocol, while the supervised FFNO control isolates one training-procedure contrast. The CNS results use 512 training trajectories in the family comparison table as secondary cross-domain architecture evidence. They show that SRU-Net is competitive in a forward-PDE setting and improves over FNO and U-Net baselines, but they do not make SRU-Net the best CNS model; FFNO is the lowest-error CNS row. The SRU-Net* notation marks the CNS spectral-bottleneck row.

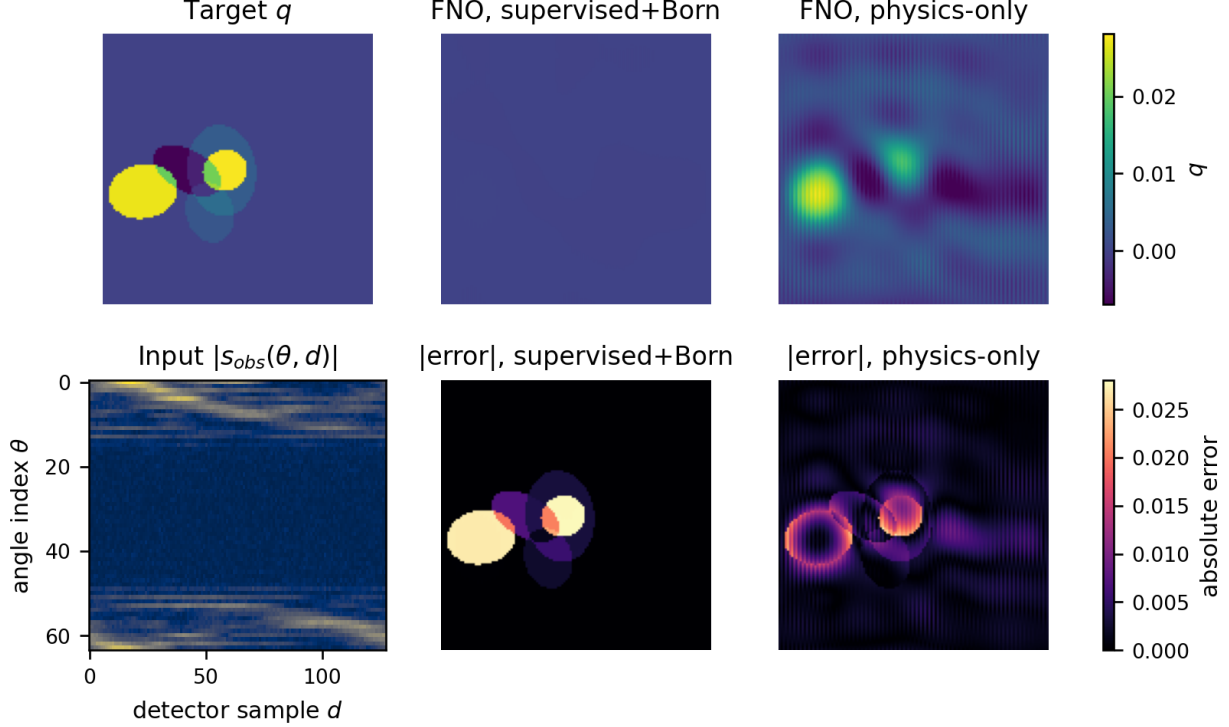


Figure 5: **BRDT FNO objective control.** Vanilla FNO did not converge under the supervised object loss plus Born-consistency objective used for the main BRDT comparison, producing a nearly constant reconstruction. Under a physics-only objective on the same BRDT split and representative example, the same architecture recovers visible object structure. This control is therefore reported separately from Fig. 4, whose columns remain a controlled comparison of the reported BRDT rows.

A BRDT FNO Objective Control

B Model Configurations

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Table 8: **Model configurations by benchmark.** Parameter counts are unique trainable parameters in the effective prediction model. Wrapper state-dict counts are not used when they duplicate the same generator under multiple aliases.

Benchmark	Model row	Training	Width	Modes	Enc.	Bott.	Down	Unique params
Synthetic CDI	CNN (pinn)	PINN	not_recorded	not_recorded	not_recorded	not_applicable	not_applicable	4,661,212
Synthetic CDI	FNO (pinn.fno.vanilla)	PINN	32	12	4	not_applicable	not_applicable	636,450
Synthetic CDI	FFNO-local proxy (pinn.ffno)	PINN	32	12	4	not_applicable	not_applicable	161,958
Synthetic CDI	SRU-Net (pinn.hybrid.resnet)	PINN	32	12	4	6	2	9,003,299
Synthetic CDI	U-NO (pinn.neuralop.uno)	PINN	32	12	4	not_applicable	not_applicable	1,257,858
Synthetic CDI	CNN (baseline)	supervised	not_recorded	not_recorded	not_recorded	not_applicable	not_applicable	4,612,418
Synthetic CDI	FFNO-local proxy (supervised.ffno)	supervised	32	12	4	not_applicable	not_applicable	161,958
Synthetic CDI	U-NO (supervised_neuralop.uno)	supervised	32	12	4	not_applicable	not_applicable	1,257,858
PDEBench CNS	FFNO (author.ffno.cns.base)	supervised mse	64	16	24	not_applicable	not_recorded	1,074,440
PDEBench CNS	SRU-Net (spectral.resnet.bottleneck.base)	supervised mse	32	12	4	6	2	8,395,270
PDEBench CNS	FNO (fno.base)	supervised mse	32	12	4	not_applicable	not_recorded	358,628
PDEBench CNS	U-Net (UNET_strong)	supervised mse	32	not_recorded	not_recorded	not_applicable	not_recorded	7,768,036
BRDT	SRU-Net (hybrid.resnet)	supervised+Born	16	8	2	2	1	142,018
BRDT	FFNO-local proxy (ffno)	supervised+Born	16	8	4	not_applicable	not_recorded	36,674

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